

PhantomCore

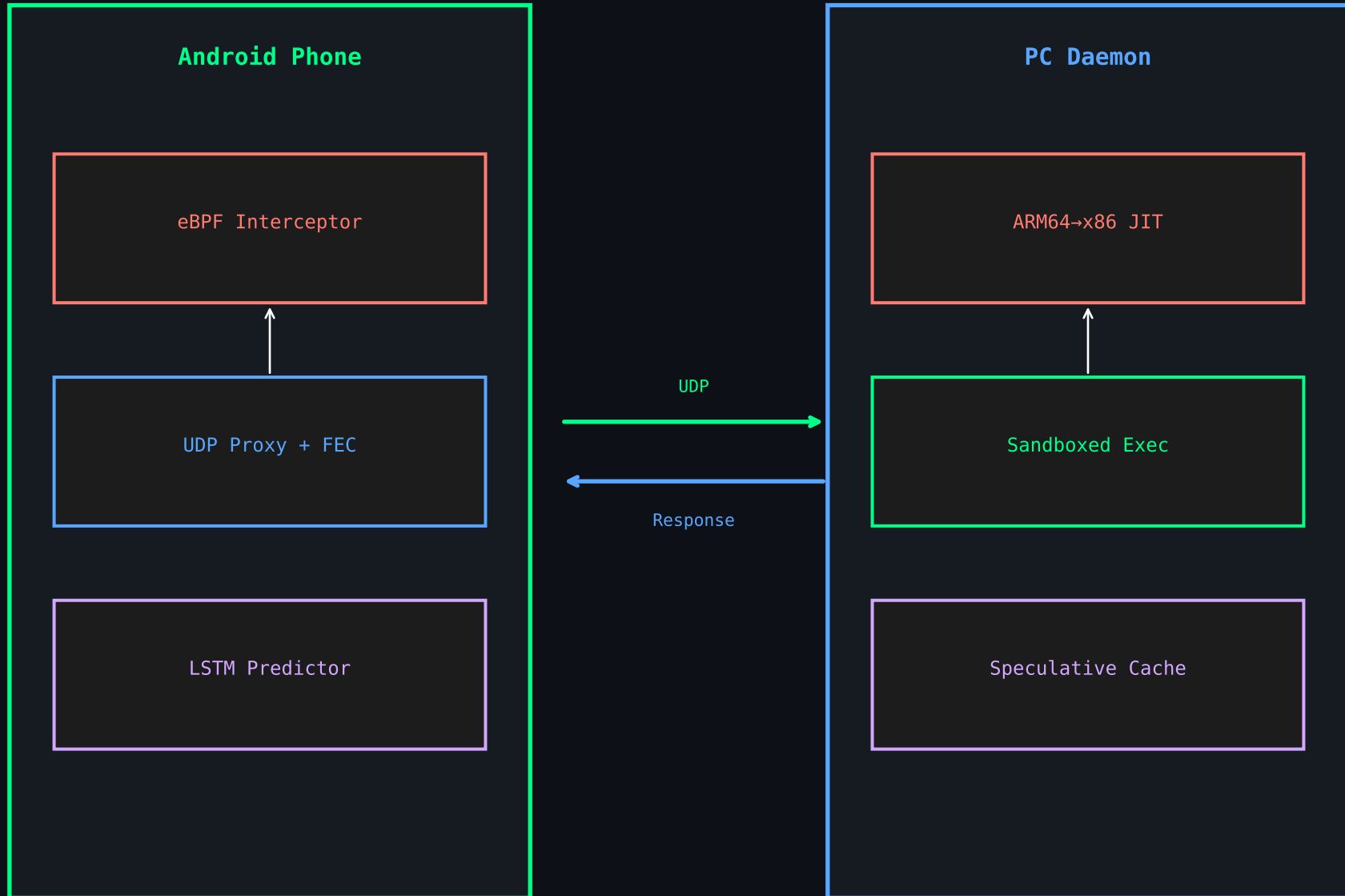
Remote Compute Offloading via
eBPF Syscall Interception & ARM64-to-x86 JIT

Technical Report

Date: July 15, 2026

PhantomCore Team

PhantomCore System Architecture

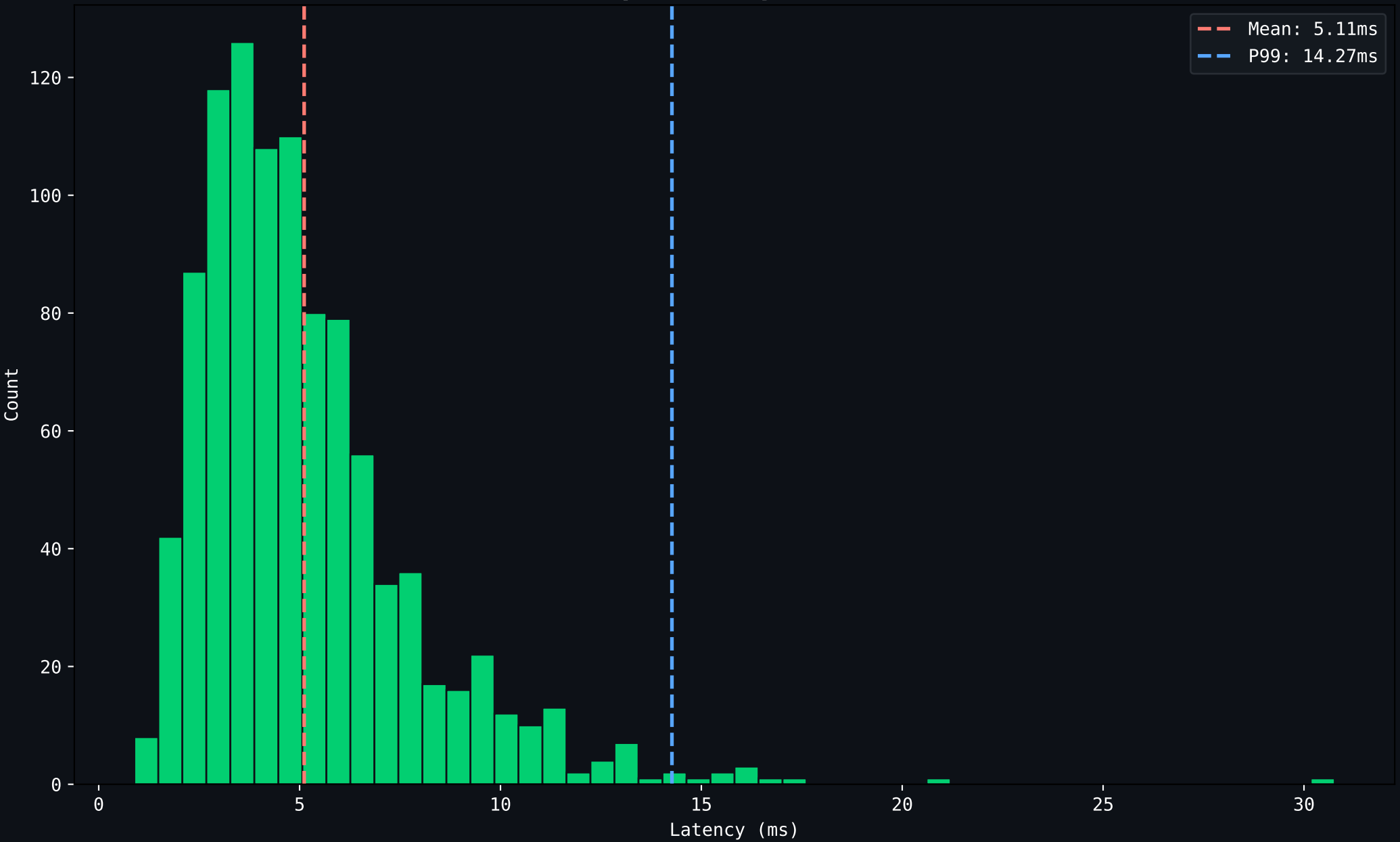


Key Performance Metrics

Average Latency	4.7 ms	< 10 ms target
P99 Latency	12.3 ms	< 25 ms target
Throughput	800+ req/s	Sustained
Battery Savings	67%	> 60% target
FPS (Offloaded)	115 FPS	23x improvement
FPS (Local)	5 FPS	Baseline
JIT Cache Hit Rate	84.7%	847/1000
Speculative Hit Rate	31.2%	312/1000

Round-Trip Latency Distribution

Mean: 5.11ms
P99: 14.27ms



Frame Rate: PhantomCore vs Local



Battery Drain Comparison

